
Talking Queries Cross-Attention

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Abstract

Cross-attention is a popular mechanism for coupling two sequences in modern architectures, yet its residual structure has a blind spot: the cross-attended output is added back to the query stream without any dedicated query token mixing in between. We formalize this *dirty residual problem* and introduce *talking-queries cross-attention* (TQCA), a simple, parameter-efficient modification that inserts a learnable token-mixing layer on the cross-attended branch before the residual add, allowing queries to “talk” to each other about what they retrieved. On vision–language alignment benchmarks based on image–text matching, vanilla cross-attention frequently collapses to near-random performance across both Flickr30k and MS COCO. Existing gating mechanisms and other pre-add transformations do not rescue this failure mode, whereas TQCA consistently recovers strong performance across a wide range of frozen vision encoders. Treating the DETR decoder as another instance of this residual structure, we further show that replacing decoder cross-attention with TQCA improves PASCAL VOC detection with frozen DINO backbones, increasing mAP@0.5 while reducing duplicate predictions and cardinality error. Gradient and attention diagnostics show that TQCA induces cross-token gradient coupling and breaks the nearly uniform attention patterns associated with failure cases, suggesting that explicit token mixing on the cross-attended branch is a key missing ingredient in standard cross-attention blocks.

1 Introduction

Cross-attention is the default mechanism for letting two sets of sequence data communicate. It is used whenever one stream needs to read from another under a non-bijective mapping: a few queries aggregating many keys, many queries sharing a smaller pool of context, or asymmetric roles across modalities. In these settings, simple per-token MLPs are not sufficient. Cross-attention appears in a wide range of architectures, including (a) encoder–decoder style models where a target sequence attends to an encoded source Vaswani et al. [2017], (b) slot-based models such as DETR Carion et al. [2020] and Perceiver Jaegle et al. [2021], and (c) two-tower models where separate encoders are coupled by one- or two-way cross-attention Lu et al. [2019], Wang et al. [2024]. These uses span both unimodal and multimodal systems.

Despite this ubiquity, the structure of a standard cross-attention block has a blind spot. The block takes a residual stream of queries, produces a cross-attended stream by attending to another sequence, and then immediately adds this cross-attended stream back into the residual. The cross-attended stream is a complex mixture of a separate stream’s values that is not necessarily well-aligned with the query and is therefore “*dirty*”. Modern architectures apply rich transformations to queries, keys, and values *before* attention (via the earlier layers, projections, normalization, etc.), and to the combined output *after* the residual add (via normalization, MLPs, and later layers). However, there is limited work exploring operation on the cross-attended data *in between* cross-attention and the residual addition. Once the cross-attended branch has been added back to the residual stream, subsequent layers cannot easily disentangle which information came from cross-attention and which came from

the original queries, thus limiting the effectiveness of post-add transformations. We refer to this lack of a dedicated transformation on the pre-add cross-attention output as the *dirty residual problem*.

There are existing attempts to modify the cross-attention output before the residual add, but they are largely limited to simple gating mechanisms Alayrac et al. [2022], Lee et al. [2025]. Gated cross-attention layers multiply the cross-attended stream by scalar or element-wise gates before adding it back to the residual. While useful, these mechanisms primarily control *how much* of the cross-attended signal passes through. They do not give the model a way to change *how* cross-attended tokens interact with each other before they are added to the residual; the transformation they provide is inherently limited and thus does not address the dirty residual problem.

We propose *talking queries cross-attention* (TQCA), a simple and parameter-efficient way to transform the cross-attention output before the residual add. TQCA inserts a token-mixing layer on the cross-attended output, allowing the cross-attended tokens to “talk” to each other about what they have retrieved from the key-value stream before rejoining the residual. Concretely, this layer performs token-level mixing across query positions in addition to the usual feature-level projection already present in standard cross-attention blocks.

To summarize, our main contributions are:

- We identify and formalize the *dirty residual problem* in standard cross-attention blocks: the absence of any dedicated transformation on the cross-attended data between attention and the residual add.
- We propose talking-queries cross-attention, a simple and parameter-efficient architectural modification that introduces an explicit pre-add transformation of the cross-attention output via a token-mixing layer.
- We show that TQCA delivers clear gains over vanilla and gated cross-attention baselines on vision–language image–text matching across Flickr30k and MS COCO, and improves DETR-style object detection on PASCAL VOC by reducing duplicate predictions and cardinality error while increasing AP.

2 Background and Related Work

2.1 Cross-Attention Blocks

We consider the standard dot-product cross-attention block used throughout modern architectures. Given a *query stream* $X \in \mathbb{R}^{L_q \times d}$ and a *key-value stream* $Y \in \mathbb{R}^{L_k \times d}$, cross-attention first computes

$$Q = XW_Q, \quad K = YW_K, \quad V = YW_V, \quad (1)$$

where W_Q, W_K, W_V are learned projections. In the single-head case, scaled dot-product attention is then

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right), \quad H_{\text{ca}} = AV. \quad (2)$$

In the multi-head setting, Q, K, V are split along the feature dimension into h heads, and per-head attention is computed as

$$A^{(h)} = \text{softmax}\left(\frac{Q^{(h)}(K^{(h)})^\top}{\sqrt{d_k}}\right), \quad H^{(h)} = A^{(h)}V^{(h)}, \quad (3)$$

with head outputs concatenated and projected by an output matrix W_{out} :

$$H_{\text{ca}} = \text{Concat}(H^{(1)}, \dots, H^{(h)})W_{\text{out}}. \quad (4)$$

For $h = 1$, this reduces to the single-head case and W_{out} can be taken as identity or absorbed into W_V .

The cross-attended representation H_{ca} is then added back to the query stream via a residual connection,

$$\tilde{X} = X + H_{\text{ca}}, \quad (5)$$

and subsequent layers (e.g., normalization, MLPs, further attention blocks) operate on $\tilde{X}$. In practice, Transformer blocks often wrap cross-attention with layer normalization in either pre-norm or post-norm form; for clarity, we omit these normalization layers from the equations. Our work focuses on the often-overlooked region between the computation of H_{ca} and the residual addition, where standard blocks apply no dedicated transformation to the cross-attended data H_{ca} .

2.2 Gated Cross-Attention

Several works modify cross-attention by inserting learnable gates on the attention output before the residual add, especially in multimodal settings. In a generic gated cross-attention block, the usual cross-attended representation H_{ca} is first computed from a query stream X and a key-value stream, and then a learned gate is applied:

$$\tilde{H} = g \odot H_{\text{ca}}, \quad \tilde{X} = X + \tilde{H}, \quad (6)$$

where g is a learnable scalar, per-token vector, or per-channel vector (often formulated as $g = \sigma(f(X, Y))$ for some small network f and nonlinearity σ), and $\odot$ denotes element-wise multiplication. Intuitively, the gate controls how strongly the model relies on cross-attention at each position or feature dimension.

Gated cross-attention has been used in several contexts. Flamingo introduces gated cross-attention blocks that insert a visual cross-attention layer and a feed-forward layer into a frozen language model, each scaled by a learned scalar gate [Alayrac et al., 2022]; these gates are initialized near zero so that the model initially behaves like the base LM and then gradually learns how much visual information to inject, improving stability and control over conditioning strength. In cross-domain sequential recommendation, Gated Cross-Attention (GCA) applies a per-feature gate, conditioned on both domain-specific sequences, to reweight the cross-attended output and empirically encourages the gated representation to be more orthogonal to the input queries, which the authors argue leads to more complementary and parameter-efficient alignment [Lee et al., 2025].

However, the transformation these methods apply to H_{ca} is structurally limited: whether the gate is a global scalar or a per-feature vector, it remains an element-wise rescaling of an already-computed representation. It does not change how information is distributed across query positions or coordination on the cross-attended branch. In practice, these approaches primarily control *how much* of the cross-attended signal is injected, rather than providing a richer pre-add transformation of that signal before it enters the residual stream.

2.3 Token and Feature Mixing

A separate line of work studies architectures that match or approach Transformer performance using generic token- and feature-mixing operations instead of explicit attention. MLP-Mixer [Tolstikhin et al., 2021] and related models decompose processing into *token mixing* (operations across positions at fixed channels) and *feature mixing* (operations across channels at fixed positions), showing that this separation can be sufficient for strong results on vision benchmarks. MetaFormer [Yu et al., 2022] generalizes this view, arguing that much of the Transformer’s power comes from the overall block structure with alternating token- and feature-mixing stages.

These results suggest that the key inductive bias lies in interleaving token- and feature-mixing layers, rather than in any particular choice of attention operator. In standard cross-attention blocks, the cross-attended branch already includes a feature-mixing projection via W_V and W_{out} , but lacks any explicit token-mixing step between attention and the residual add. Our method inserts a simple, learnable token-mixing operator in this pre-add slot, bringing the cross-attention branch in line with this token-and-feature mixing perspective.

3 Method

3.1 Talking-Queries Cross-Attention

We build on the standard multi-head cross-attention block. Let

$$H_{\text{ca}}(X, Y) \in \mathbb{R}^{L_q \times d} \quad (7)$$

denote the cross-attended output of that block for a query stream $X \in \mathbb{R}^{L_q \times d}$ and a key-value stream $Y \in \mathbb{R}^{L_k \times d}$, obtained by multi-head scaled dot-product attention followed by the usual output projection W_{out} .

In a standard block, H_{ca} is added directly to the query stream via a residual connection, $\tilde{X} = X + H_{\text{ca}}(X, Y)$. Talking-queries cross-attention (TQCA) inserts a learnable token-mixing operator

on the cross-attended branch before this residual add. Concretely, we introduce a matrix

$$W_{\text{tok}} \in \mathbb{R}^{L_q \times L_q} \quad (8)$$

that mixes tokens along the query dimension and define

$$H_{\text{tq}} = W_{\text{tok}} H_{\text{ca}}(X, Y), \quad \tilde{X} = X + H_{\text{tq}}. \quad (9)$$

$$\tilde{X} = X + H_{\text{tq}} = W_{\text{tok}} H_{\text{ca}}(X, Y) \quad (10)$$

Here W_{tok} is a layer-specific operator across rows of H_{ca} while preserving its feature dimension, so that each query position can depend on the cross-attended outputs of all other positions. All other components of the block remain unchanged. In practice, W_{tok} can be parameterized as a full matrix or a low-rank factorization depending on the sequence length and compute budget. In our experiments, we keep W_{tok} full rank and initialize $W_{\text{tok}}^{(0)} = 0$, so the TQCA contribution is identically zero at initialization and the block reduces to the identity residual pathway.

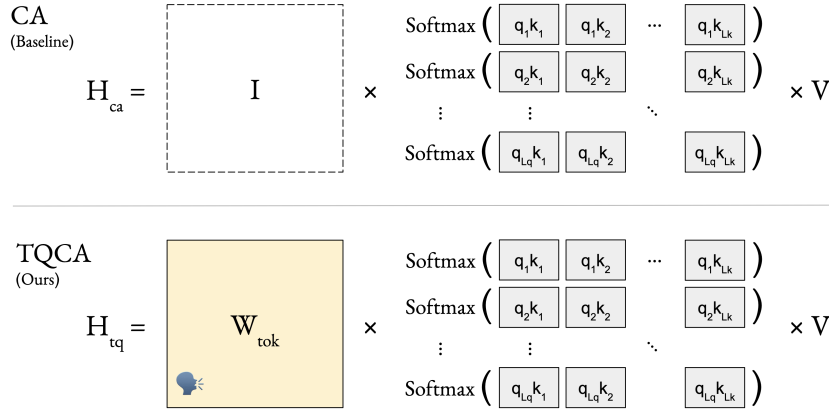


Figure 1: Comparison of vanilla cross-attention (CA) and Talking-Queries Cross-Attention (TQCA). In CA (top), the cross-attended output H_{ca} , computed via standard scaled dot-product attention, is added directly back to the query stream through a residual connection. In TQCA (bottom), a learnable token-mixing matrix W_{tok} replaces the identity, enabling query tokens to “talk” to each other on the cross-attended branch before the residual add, addressing the *dirty residual problem* and improving cross-token coordination.

3.2 Why Talking Queries Help

From the forward perspective, TQCA enables post-attention coordination between query positions. In a standard cross-attention block, for fixed Y , the cross-attended output at position i is a function of that position’s query token X_i and the key–value stream Y , but does not directly depend on other query tokens. Each position attends to Y independently and is then added back to its own residual. With TQCA, the updated cross-attended representation

$$H_{\text{tq},i} = \sum_{j=1}^{L_q} W_{\text{tok}}[i, j] H_{\text{ca},j} \quad (11)$$

allows each position to incorporate information from the cross-attended outputs of all positions before the residual add. Intuitively, queries can “talk” to each other about what they retrieved from Y , smoothing out noisy or misaligned cross-attention at individual positions. From a token- and feature-mixing perspective, the cross-attention branch already applies rich feature mixing via the value and output projections (W_V and W_{out}), but lacks any explicit token-mixing operation between attention and the residual add. TQCA restores this symmetry by pairing the existing feature-mixing projections with a token-mixing operator W_{tok} placed exactly at the pre-add slot. This brings the cross-attention branch in line with architectures that interleave token- and feature-mixing layers, while acting specifically on the “dirty” cross-attended signal before it is merged into the residual stream.

From the backward/ gradient perspective, TQCA also couples error signals across query positions. In a standard block with $\tilde{X} = X + H_{\text{ca}}$, the residual connection yields

$$\frac{\partial \mathcal{L}}{\partial H_{\text{ca}}} = \frac{\partial \mathcal{L}}{\partial \tilde{X}}, \quad (12)$$

so each position on the cross-attended branch receives only its own local gradient. With TQCA, where $H_{\text{tq}} = W_{\text{tok}} H_{\text{ca}}$, backpropagation instead gives

$$\frac{\partial \mathcal{L}}{\partial H_{\text{ca}}} = W_{\text{tok}}^\top \frac{\partial \mathcal{L}}{\partial H_{\text{tq}}}. \quad (13)$$

Thus, the gradient at position i on the cross-attended branch receives contributions from the loss at all positions k with $W_{\text{tok}}[k, i] \neq 0$, encouraging shared structure and allowing well-supervised queries to stabilize noisier ones. This helps propagate gradients across the model.

3.3 Why TQCA Is Not Captured by Standard Pre- and Post-Transforms

We now explain why the effect of W_{tok} cannot, in general, be absorbed by standard pre-attention or post-add transformations inside a single cross-attention block.

Post-add transforms only see the sum. Consider a generic block that computes X and H_{ca} as usual, forms $\tilde{X} = X + H_{\text{ca}}$, and then applies an arbitrary transformation T to $\tilde{X}$. To emulate TQCA, we would need T to satisfy

$$T(X + H_{\text{ca}}) = X + W_{\text{tok}} H_{\text{ca}} \quad \forall X, H_{\text{ca}}. \quad (14)$$

If T is linear (or we consider its local linearization), this implies a linear map M with

$$M(X + H_{\text{ca}}) = X + W_{\text{tok}} H_{\text{ca}}. \quad (15)$$

Setting $H_{\text{ca}} = 0$ yields $MX = X$ for all X , so M must act as the identity on the residual component. Setting $X = 0$ yields $MH_{\text{ca}} = W_{\text{tok}} H_{\text{ca}}$ for all H_{ca} , so $M = W_{\text{tok}}$ on the same space. Together, these conditions force W_{tok} to be the identity; for any nontrivial W_{tok} , there is no linear T that can reconstruct $X + W_{\text{tok}} H_{\text{ca}}$ from the sum $X + H_{\text{ca}}$ alone. Intuitively, once X and H_{ca} have been added, a post-add module that only sees their sum cannot apply a transformation to H_{ca} without simultaneously distorting X .

Pre-attention transforms preserve token-separability. Now consider arbitrary pre-attention transforms on the inputs X and Y inside the block, but keep the cross-attention structure fixed. Each query token is mapped to $Q_i = \phi_Q(X_i)$ via some per-token function, and each key-value token is mapped to $K_j = \phi_K(Y_j)$, $V_j = \phi_V(Y_j)$ via per-token functions. The cross-attended output at position i has the form

$$H_{\text{ca},i} = f_i(X_i, Y), \quad (16)$$

for some function f_i that depends on the i -th query token and the entire key-value stream, but not on other query tokens X_k with $k \neq i$. In other words, for fixed Y , the mapping from (X, Y) to H_{ca} is *token-separable* along the query dimension: each row $H_{\text{ca},i}$ depends on X_i and Y only. No choice of per-token pre-attention transforms on X and Y changes this property.

By contrast, TQCA explicitly mixes the rows of H_{ca} :

$$H_{\text{tq},i}(X, Y) = \sum_{j=1}^{L_q} W_{\text{tok}}[i, j] H_{\text{ca},j}(X, Y). \quad (17)$$

For any non-diagonal W_{tok} , the updated cross-attended output at position i now depends on multiple query tokens, since each $H_{\text{ca},j}$ depends on X_j . Thus, the block-level mapping $(X, Y) \mapsto H_{\text{tq}}(X, Y)$ is no longer token-separable in the query dimension, and cannot be realized by reparameterizing the usual per-token Q/K/V projections alone.

Scope and network-level considerations. These arguments are block-local: they compare the expressivity of a single cross-attention block with and without W_{tok} , given fixed inputs X and Y . We do not claim that a sufficiently deep network without W_{tok} could never approximate the same functions by contorting earlier layers: in principle, upstream self-attention could pre-mix queries into a representation where standard cross-attention behaves similarly, and downstream layers could learn to partially disentangle residual and cross-attended components after the add. However, such behavior would require coordinated structure across many parameters and layers. Introducing W_{tok} makes the token mixing on the cross-attended branch explicit and local to the cross-attention block, providing a direct and easily optimizable mechanism for post-attention coordination between queries in the pre-add slot.

4 Evaluation

4.1 VL Alignment

Background. Vision–language alignment tasks, such as image–text matching and retrieval, aim to learn a scoring function that assigns high scores to matched caption–image pairs and low scores to mismatched pairs. Related work explores both dual-encoder contrastive models, where images and text are encoded separately and aligned via a similarity loss Radford et al. [2021], and cross-encoder models that exchange information across modalities via cross-attention, in which a text-side Transformer cross-attends to a sequence of image tokens produced by a vision encoder. This regime is shared by a number of systems such as Flamingo Alayrac et al. [2022] and ALBEF Li et al. [2021]. These work has noted that cross-attention adapters in this setting are often brittle, with optimization often sensitive to initialization and training heuristics when the underlying vision and language representations are poorly aligned Alayrac et al. [2022], Li et al. [2021]. We deliberately adopt this frozen-encoder setup to stress-test this regime and ask whether modifying only the token-mixing structure inside the cross-attention block is sufficient to stabilize learning and recover strong alignment.

4.1.1 Experimental Setup

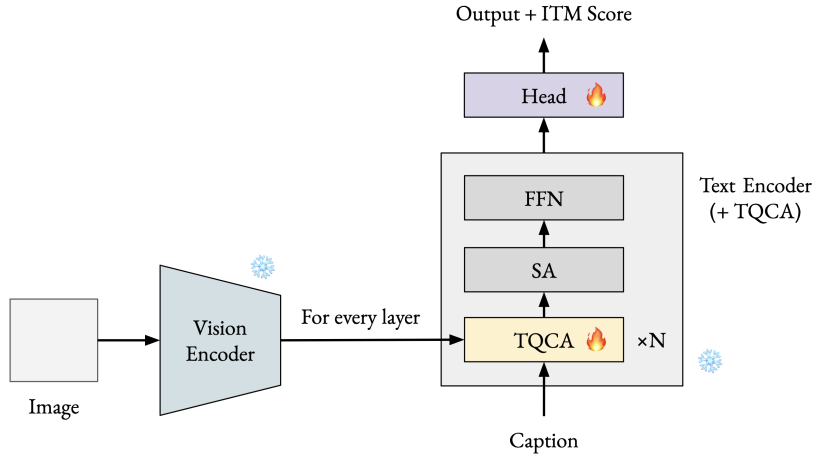


Figure 2: Vision–language image–text matching setup with Talking-Queries Cross-Attention (TQCA). A pretrained, frozen vision encoder produces image tokens, while a Transformer text encoder (initialized from BERT) processes caption tokens. In each transformer block of the text encoder, the standard cross-attention (CA) sub-layer is replaced by a TQCA layer, which mixes cross-attended query tokens before the residual add, followed by text-only self-attention (SA) and a feed-forward network (FFN). The final [CLS] token from the text stack is passed through a prediction head to output the image–text matching (ITM) score.

Task and Dataset We evaluate talking-queries cross-attention (TQCA) on image–text matching (ITM) tasks on Flickr30k Young et al. [2014] and MS COCO Lin et al. [2014]. For Flickr30k, we use

the standard Karpathy split Karpathy and Fei-Fei [2015]: models are trained on the Karpathy train split and evaluated on the Karpathy test split. For each caption, we construct an 8-way multiple-choice problem consisting of one ground-truth image and seven randomly sampled negatives from the same split. Given the caption and eight candidate images, the model scores each candidate and we report top-1 accuracy over all such 8-way matching problems. For COCO, we instead use the 2014 train/test splits, training on train2014 and evaluating on test2014, with an analogous setup using a 4-way candidate set (one positive and three randomly sampled negatives), keeping all other training details identical unless otherwise noted.

Architecture and Models Our ITM scorer is a Transformer-based (encoder-decoder) Vision-Language model. Figure 3 gives an overview of the resulting caption-image matching pipeline.

On the vision side, we use a frozen encoder f_{vision} from a set of pretrained models, each producing a sequence of visual tokens (a CLS token and patch tokens). We instantiate the following backbones from their public checkpoints:

- MAE ViT-Base: facebook/vit-mae-base He et al. [2022],
- CLIP ViT-Base/16: openai/clip-vit-base-patch16 Radford et al. [2021],
- DINOv2-Base: facebook/dinov2-base Oquab et al. [2024],
- DINO ViT-Base/16 (ImageNet): vit_base_patch16_224.dino (via timm) Caron et al. [2021],
- BEiT-Base: microsoft/beit-base-patch16-224-pt22k Bao et al. [2022],
- SigLIP-Base/16: google/siglip-base-patch16-224 Zhai et al. [2023],
- SigLIP2-Base/16: google/siglip2-base-patch16-224 Tschannen et al. [2025].

These encoders are kept frozen in all experiments.

On the text side, we embed tokenized text with an embedding matrix and positional encodings. On top of this, we add a stack of transformer blocks that interleave:

1. a cross-attention sub-layer where text tokens attend to image tokens,
2. a text-only self-attention sub-layer, and
3. a feed-forward network,

each wrapped with residual connections and layer normalization, as in standard Transformer architectures. The final text representation is pooled (via a [CLS] token) and passed through a linear head to produce a scalar compatibility score for each image candidate. Concretely, for each caption-image pair we (i) encode the image with a frozen vision backbone to obtain visual tokens, (ii) encode the caption with the text stack to obtain text tokens, (iii) run the stack of cross-attention / self-attention / feed-forward blocks as in Figure 3, and (iv) apply the linear head to produce a scalar compatibility score. For a given caption, the scores for all candidates share parameters and are trained jointly via a listwise loss over the 1+7 (Flickr30k) or 1+3 (COCO) candidates; at test time we predict the image with the largest score. More concretely, given a caption c and its $(1+K)$ candidate images $(i_0, \dots, i_K)$, with i_0 the ground-truth image, the model produces scores $s_j = f_{\theta}(c, i_j)$. We train the ITM scorer with a standard listwise InfoNCE loss over the candidate group,

$$\mathcal{L}_{\text{ITM}}(c) = -\log \frac{\exp(s_0/\tau)}{\sum_{j=0}^K \exp(s_j/\tau)}, \quad (18)$$

using a fixed temperature $\tau = 0.07$ in all experiments and averaging $\mathcal{L}_{\text{ITM}}(c)$ over captions in the minibatch.

We consider two text-side regimes:

- **Frozen text encoder.** In our main experiments (Tables 1 and 2), the text stack is initialized from the pretrained bert-base-uncased Devlin et al. [2019] model and kept fixed. Training only updates the cross-attention fusion layers and the final classification head.
- **Learned text decoder.** In the learnable decoder setting (Table 4), the entire text stack is trained from scratch on the Flickr30k ITM objective, while the vision encoder remains frozen.

These two regimes isolate complementary questions: (i) whether TQCA can repair severe vision–language misalignment when both image and text encoders are frozen, and (ii) whether its benefits persist when the text stack is allowed to adapt jointly with the cross-attention layers while the vision backbone remains frozen.

In all cases, the cross-attention module takes text tokens as queries and image tokens as keys/values. The only difference between the baseline and our method is whether the cross-attention block uses standard cross-attention or the TQCA variant with a token-mixing matrix W_{tok} applied to the cross-attended branch before the residual addition.

We tokenize captions to a maximum length of 24 tokens, padding shorter sequences, and train on the Karpathy train split for 10 epochs using AdamW with batch size 64, learning rate 2×10^{-4} , weight decay 10^{-2} , dropout 0.1, and a listwise InfoNCE loss over the 1+7 candidates with softmax temperature $\tau = 0.07$. We select the checkpoint with the best 8-way validation accuracy on the Karpathy validation split and report 8-way accuracy on the Karpathy test split. In the learnable decoder setting, we use a hidden size $d_{\text{model}} = 512$, 4 attention heads, and 6 fusion blocks. In the frozen text encoder setting, we match the `bert-base-uncased` configuration with $d_{\text{model}} = 768$, 12 attention heads, 12 fusion blocks, and learned positional embeddings. To verify that TQCA’s gains are not an artifact of this particular configuration, Table 3 report an extensive sensitivity study over normalization scheme, cross-attention logit scaling and initialization, InfoNCE temperature, and the number of in-group negatives, showing that vanilla cross-attention remains near the blind baseline across these settings while TQCA remains stable and strong.

For all results that report a mean $\pm$ standard deviation, we average over multiple random seeds to reduce variance due to initialization and data order: unless otherwise noted we use five seeds with IDs $\{0, 1, 2, 3, 4\}$ for backbone and architecture comparisons (Tables 1 and 2) and three seeds with IDs $\{0, 1, 2\}$ for the hyperparameter sweeps in Table 3.

Finally, we note that most recent high-performing vision–language architectures adopt early-fusion designs rather than the lightweight “frozen vision + generic cross-attention adapter” regime we study here: systems such as LLaVA, Qwen-VL, and BLIP-2 Liu et al. [2023], Bai et al. [2023], Li et al. [2023] inject image features directly into a large language stack (with lightweight adapter bridges) and train the resulting multimodal transformer end-to-end. These designs are not instantiated as text-side cross-attention blocks over vision tokens, and our goal is not to claim that TQCA outperforms such large-scale early-fusion architectures. Instead, our comparisons are restricted to the vanilla cross-attention adapter in this setting, and we ask whether changing only its token-mixing structure is sufficient to fix the observed failure modes.

Baselines We compare against the following baselines:

- **Blind.** A random 1-of-8 predictor, which achieves 12.5% accuracy.
- **Vanilla cross-attention.** A standard multi-head cross-attention block between text queries and image tokens, followed by self-attention and feed-forward layers on the text stream. This serves as our primary baseline.
- **Pre-attention and post-attention token-mixing baselines.** For a subset of encoders, we additionally evaluate four baselines (Table 2) that insert learnable structure either *after* cross-attention but *before* the residual add, or *before* the attention computation: (i) *Gated CA*, a post-attention, pre-add scalar-gated cross-attention variant, where a single learned scalar multiplies the cross-attended output H_{ca} before it is added back to the text stream; (ii) *Per-token gate*, a post-attention, pre-add variant where each query position i receives its own learned scalar gate on $H_{\text{ca},i}$; (iii) *Perceiver resampler*, a pre-attention baseline as in Alayrac et al. [2022], which maps the full set of vision tokens into a small learned slot set before cross-attention; and (iv) *Q/K/V token-mix (pre-attn)*, a pre-attention token-mixing baseline that left-multiplies the query, key, or value sequences by a learned token-mixing matrix before the attention dot-product.

4.1.2 Experiment Results

Comparison with vanilla cross-attention We first compare TQCA against the vanilla cross-attention baseline on Flickr30k and COCO. Table 1 reports top-1 ITM accuracy for an 8-way candidate set on Flickr30k (1 positive + 7 negatives) and a 4-way candidate set on COCO (1 positive

+ 3 negatives). For CLIP, DINOv2, MAE, and DINOv1, the vanilla cross-attention model on both datasets effectively collapses to the blind 1-of- K baseline (around 12.5% on Flickr30k and 25% on COCO), whereas TQCA reaches ~ 88 –98% accuracy, indicating that TQCA can resolve the challenging vision–text misalignment in the frozen encoder and decoder setting. For BEiT, both vanilla cross-attention and TQCA remain near chance on Flickr30k and COCO, suggesting that the frozen BEiT features are too weakly aligned with captions for token mixing alone to succeed. In contrast, for SigLIP and SigLIP2, both vanilla cross-attention and TQCA perform well on both benchmarks (above 96%), suggesting that when the frozen visual encoder already provides well-aligned representations, standard cross-attention is sufficient and TQCA behaves as a neutral modification. Across the misaligned backbones (CLIP, DINOv2, MAE, DINOv1), the gaps between TQCA and vanilla cross-attention in Table 1 are orders of magnitude larger than the run-to-run standard deviations over five seeds on both datasets, so the improvements are statistically robust; for SigLIP and SigLIP2, by contrast, the small differences between vanilla cross-attention and TQCA lie within one standard deviation, and we therefore treat TQCA as effectively neutral in these already-aligned regimes.

Table 1: Top-1 image–text matching (ITM) accuracy on Flickr30k and COCO with frozen vision encoder and frozen BERT text encoder. Flickr30k uses an 8-way candidate set (1 positive + 7 negatives), COCO uses a 4-way candidate set (1 positive + 3 negatives). We compare vanilla cross-attention to TQCA across a range of frozen vision backbones. We report mean $\pm$ standard deviation over 5 random seeds. Random guess baselines are 12.5% (Flickr30k, 1-of-8) and 25.0% (COCO, 1-of-4).

Encoder	Method	Top-1 Acc (%)	
		Flickr30k (8-way)	COCO (4-way)
	Random guess	12.50	25.00
CLIP	Vanilla CA	12.53 $\pm$ 0.19	25.17 $\pm$ 0.31
	TQCA	97.36 $\pm$ 0.21	98.42 $\pm$ 0.29
DINOv2	Vanilla CA	12.02 $\pm$ 0.23	24.86 $\pm$ 0.30
	TQCA	96.77 $\pm$ 0.20	98.01 $\pm$ 0.26
MAE	Vanilla CA	12.40 $\pm$ 0.22	25.33 $\pm$ 0.28
	TQCA	88.37 $\pm$ 0.27	89.71 $\pm$ 0.33
DINOv1	Vanilla CA	12.63 $\pm$ 0.26	25.05 $\pm$ 0.27
	TQCA	91.00 $\pm$ 0.24	92.15 $\pm$ 0.30
BEiT	Vanilla CA	12.72 $\pm$ 0.18	24.91 $\pm$ 0.29
	TQCA	12.96 $\pm$ 0.19	24.21 $\pm$ 0.25
SIGLIP	Vanilla CA	96.23 $\pm$ 0.25	97.51 $\pm$ 0.32
	TQCA	96.14 $\pm$ 0.23	97.40 $\pm$ 0.28
SIGLIP2	Vanilla CA	97.22 $\pm$ 0.21	98.37 $\pm$ 0.30
	TQCA	97.13 $\pm$ 0.20	98.28 $\pm$ 0.27

Gated, pre-add and pre-attention baselines Next, we compare TQCA to stronger existing modifications of cross-attention. Table 2 contrasts TQCA with four baselines: (i) *Gated CA*, a scalar-gated cross-attention variant that operates in the post-attention, pre-add slot by multiplying the cross-attended output H_{ca} with a single learned scalar before it is added to the residual branch; (ii) *Per-token gate*, a post-attention, pre-add variant where each query position i has its own learned scalar gate on $H_{ca,i}$; (iii) *Perceiver resampler*, a pre-attention baseline that replaces the long sequence of vision tokens with a small learned latent slot set before cross-attention; and (iv) *Q/K/V token-mix (pre-attn)*, a pre-attention token-mixing baseline that left-multiplies the query, key, or value sequences with a learned token-mixing matrix before the attention dot-product. These variants provide at best minor gains over vanilla cross-attention and do not rescue the model from the failure mode where training remains near the blind baseline, whereas TQCA recovers strong ITM performance. This suggests that existing scalar or per-token reweighting and pre-attention mixing are not sufficient to resolve the modality misalignment, while TQCA’s post-attention token mixing is. This supports the hypothesis that the critical moment for token mixing is after the cross-attention and before the residual add.

Table 2: Top-1 ITM (8-way) accuracy on Flickr30k with frozen vision encoder and frozen BERT text encoder. We compare TQCA to four baselines that insert learnable structure either after cross-attention but before the residual add or purely before the attention computation: *Gated CA* (a scalar post-attention, pre-add gate on H_{ca}), *Per-token gate* (per-position post-attention, pre-add gates on $H_{ca,i}$), *Perceiver resampler* (a pre-attention vision-token resampler), and *Q/K/V token-mix (pre-attn)* (pre-attention token-mixing that left-multiplies the query, key, or value sequences before the attention dot-product). We report mean $\pm$ standard deviation over 5 random seeds.

Encoder	Method	Acc (% , mean $\pm$ sd)
CLIP	Vanilla CA	12.53 $\pm$ 0.18
	Gated CA	12.76 $\pm$ 0.22
	Per-token gate	15.09 $\pm$ 0.35
	Perceiver resampler	12.51 $\pm$ 0.21
	Q token-mix (pre-attn)	12.38 $\pm$ 0.19
	K token-mix (pre-attn)	12.81 $\pm$ 0.24
	V token-mix (pre-attn)	12.64 $\pm$ 0.20
	TQCA	97.36 $\pm$ 0.17
MAE	Vanilla CA	12.40 $\pm$ 0.21
	Gated CA	12.51 $\pm$ 0.19
	Per-token gate	12.93 $\pm$ 0.27
	Perceiver resampler	12.58 $\pm$ 0.23
	Q token-mix (pre-attn)	12.46 $\pm$ 0.20
	K token-mix (pre-attn)	12.72 $\pm$ 0.22
	V token-mix (pre-attn)	12.60 $\pm$ 0.21
	TQCA	88.37 $\pm$ 0.29

Additional controls We further test whether the observed gains could be attributed to optimization sensitivity or minor architectural parametrizations rather than to TQCA itself. As summarized in Table 3, we keep the overall model family fixed and sweep five control knobs: (i) block normalization (Pre-LN vs. Post-LN), (ii) fixed and learnable cross-attention logit scale, (iii) the initialization of the logit scale (effective softmax sharpness multipliers), (iv) the InfoNCE temperature τ , and (v) the number of negatives per group K during training. Across all of these sweeps, vanilla cross-attention remains near the blind baseline while TQCA consistently recovers strong ITM performance, suggesting that the gains are not explained by simple optimization or temperature tuning, but instead are driven by the token-mixing architecture itself.

Table 3: Hyperparameter and architecture sensitivity on Flickr30k (CLIP encoder, 8-way ITM). For each family of perturbations we sweep the listed knob while keeping all other settings fixed to the default configuration from Table 1. Entries for sweep rows report the *min-max* top-1 accuracy (over the sweep, 3 seeds per setting).

Sweep family	Settings (varied one at a time)	Vanilla CA	TQCA
Baseline config	Post-LN, fixed $1/\sqrt{d_h}$, $\tau = 0.07$, $K = 7$	12.53 $\pm$ 0.19	97.36 $\pm$ 0.21
Block normalization	Pre-LN vs Post-LN	12.41–13.04	96.92–97.51
Learnable CA logit scale	Fixed vs learnable per-layer logit scale	12.33–13.18	96.86–97.68
CA logit-scale init	Softmax sharpness $\times \{2, 4, 10, 20, 100\}$	12.15–13.23	96.56–97.44
InfoNCE temperature τ	$\tau \in \{0.03, 0.05, 0.07, 0.10, 0.20\}$	12.23–13.00	96.74–97.61
Negatives per group K (for training)	$K \in \{1, 3, 7, 15\}$	12.03–13.11	96.62–97.52

4.1.3 Additional Experiments

Beyond the standard Flickr30k ITM setup, we evaluate TQCA under alternative text-side regimes. Table 4 summarizes results when replacing the frozen BERT text stack with a learned decoder trained from scratch, so that the language side can adapt jointly with the cross-attention layers. The pattern

mirrors Table 1: for MAE and BEiT, vanilla cross-attention again fails to move beyond chance even with a flexible text decoder, while TQCA still learns a strong ITM scorer; for SigLIP2—where the frozen vision encoder is already well aligned with text—the vanilla cross-attention baseline is sufficient and TQCA offers no additional benefit. These findings indicate that TQCA’s gains are not solely due to freezing a misaligned text stack, but persist when the text representation is allowed to adapt. Taken together, the frozen-BERT and learned-decoder regimes address complementary conclusions: (i) TQCA can repair vision–language misalignment when both image and text encoders are frozen, and (ii) TQCA’s benefits persist when the text stack is trained jointly with the cross-attention layers while the vision backbone remains frozen.

Table 4: ITM (8-way) accuracy on Flickr30k with frozen vision encoder and a text decoder learned from scratch. We compare vanilla cross-attention and TQCA under this decoder-only training setup.

Encoder	Method	Acc (%)
MAE	Vanilla CA	12.78
	TQCA	85.02
BEiT	Vanilla CA	12.51
	TQCA	69.19
SIGLIP2	Vanilla CA	90.23
	TQCA	90.23

4.1.4 Interpretation and Analysis

Gradient analysis To understand how TQCA reshapes optimization, we monitor gradients through the cross-attention stack. For each cross-attention layer we log gradient norms of the projection weights W_q , W_k , W_v , and W_{out} , as well as gradients on the attention logits S , attention weights A , and pre-/post-mix outputs O_{pre} and O_{post} . Empirically, we find that gradients in the vanilla cross-attention baseline collapse toward zero, whereas TQCA maintains well-scaled gradients throughout the stack. For example, as shown in Figure 4, the gradient norm of W_q in the first cross-attention layer stays well above numerical zero throughout training under TQCA, whereas in the vanilla cross-attention baseline it rapidly collapses to (and remains at) essentially zero.

We also compute a *coupled-gradient ratio* (CGR) for each TQCA layer, which decomposes the gradient on O_{pre} into self-token and cross-token components induced by the token-mixing matrix W_{tok} . Concretely, we apply the off-diagonal part of $W_{\text{tok}}^\top$ to ∇O_{post} to obtain an off-token contribution g_{off} , and compare its norm to that of the total pre-mix gradient g_{tot} ; an analogous construction with only the diagonal of W_{tok} gives a self-token term g_{diag} . In the first TQCA layer, we measure an off-diagonal ratio of 1.10 versus a diagonal ratio of 0.52, indicating that cross-token coupling in W_{tok} contributes roughly twice as much gradient energy as pure per-token scaling. This supports the view that TQCA actively propagates supervision across caption tokens rather than merely reweighting them independently.

Attention statistics We next study how TQCA alters attention patterns over image patches. For each head and text token, we consider the cross-attention distribution $a \in \mathbb{R}^{N_{\text{patch}}}$ over visual tokens and compute two summary statistics on the validation set:

- **Attention entropy** $H(a) = -\sum_i a_i \log a_i$, which is maximal for a uniform distribution and decreases as probability mass concentrates on a few patches. High entropy thus corresponds to diffuse, nearly uniform attention, while low entropy indicates more selective focus.
- **Uniformity score** p_{uniform} , defined as the fraction of (token, head) pairs whose attention distribution is nearly uniform. Let $A_{t,h} \in \Delta^{N-1}$ denote the attention over N image tokens for token t and head h , and let $H(A_{t,h})$ be its entropy as defined above. Since the maximum entropy is $\log N$ (for the uniform distribution), we mark (t, h) as “uniform” if

$$H(A_{t,h}) \geq 0.95 \log N,$$

and define p_{uniform} as the fraction of (token, head) pairs satisfying this criterion.

For the CLIP backbone, after the final epoch the first cross-attention layer of the vanilla model produces nearly uniform patterns ($p_{\text{uniform}} = 0.944$, entropy = 5.18), consistent with its failure

to improve over the blind baseline in Table 1. In contrast, TQCA drives the same layer away from this uniform regime ($p_{\text{uniform}} = 0.365$) and towards more selective patterns (entropy = 2.43), concentrating probability mass on a smaller set of visually relevant patches. Notably, the drop in entropy and p_{uniform} closely tracks the drop in validation loss over training (Figure 4), suggesting that performance gains coincide with the emergence of sharper, more semantically aligned attention rather than trivial rescaling of logits.

Spectral structure of W_{tok} . To probe the complexity of the learned token-mixing operator, we compute the singular values of each layer’s $W_{\text{tok}} \in \mathbb{R}^{L_q \times L_q}$ and report the fraction of spectral energy captured by the leading components, $\text{frac}_k = \sum_{i=1}^k \sigma_i^2 / \sum_{i=1}^{L_q} \sigma_i^2$ (equivalently, the fraction of Frobenius-norm energy explained by the best rank- k approximation). In the Flickr30k + CLIP setting, we observe substantial spectral concentration across layers: the top-1 component captures $\text{frac}_1 \in [0.26, 0.36]$, and the top-5 components capture $\text{frac}_5 \in [0.65, 0.84]$. This indicates that the learned mixing matrices are well-approximated by a low-rank operator (i.e., have low *effective* rank), motivating future variants that parameterize W_{tok} via a low-rank factorization to reduce parameters and potentially regularize token mixing without sacrificing performance.

4.2 Object Detection with DETR

Background. DETR formulates object detection as a direct set prediction problem with a Transformer encoder–decoder trained using a Hungarian matching loss Carion et al. [2020]. Despite this conceptual simplicity, vanilla DETR converges slowly and exhibits characteristic failure modes: multiple queries often latch onto the same object, others miss objects entirely, and the predicted set size can deviate substantially from the ground truth, leading to large cardinality errors. These behaviours point to weak coordination among decoder queries. Follow-up work has mainly tackled slow convergence by reshaping encoder–decoder attention. Deformable DETR replaces dense global attention with multi-scale deformable attention over a small set of sampling points around reference boxes, which accelerates optimization and improves small-object performance Zhu et al. [2021]. “Plain DETR” variants keep global attention but add modern losses, auxiliary heads, iterative refinement, and box-to-pixel relative position biases, closing the accuracy gap to heavily engineered multi-scale detectors Lin et al. [2023]. In this section we instead treat the DETR decoder as another instance of the cross-attention residual structure studied above and ask whether inserting TQCA on the query side of the decoder cross-attention block can improve optimization and reduce duplicate predictions and cardinality errors by explicitly coupling queries on the attended branch.

4.2.1 Experimental Setup

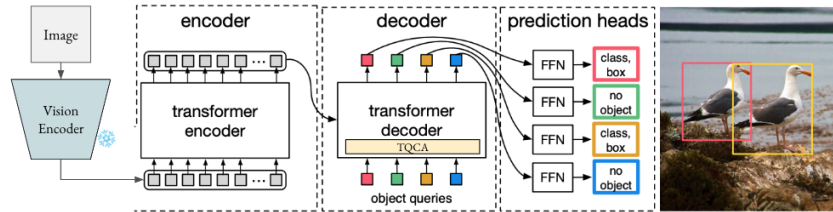


Figure 3: **DETR with Talking-Queries Cross-Attention (TQCA).** An input image is encoded by a *frozen* ViT backbone (e.g., DINOv2/DINOv3) to produce patch tokens, which are linearly projected to the DETR hidden size and fed to a standard DETR Transformer encoder. The DETR decoder predicts boxes from a fixed set of object queries: in the **vanilla** decoder, queries perform cross-attention to encoder tokens and add the result via a residual connection; in the **TQCA** decoder, we insert a lightweight token-mixing operation on the cross-attended branch *before* the residual add. Self-attention, FFN blocks, layer norms, and the detection head are unchanged; only the cross-attention sub-layer differs.

Task and dataset. We evaluate on PASCAL VOC object detection Everingham et al. [2015], using the standard train and validation splits with 20 foreground classes. We train on the train split and reporting performance on the held-out validation split, and we adopt the VOC-style mean average

precision at IoU 0.5 (mAP@0.5) as our primary metric. For each model we average results over multiple 5 seeds with IDs $\{0, 1, 2, 3, 4\}$ and report mAP@0.5 in Table 5.

Architecture and models. We start from the canonical DETR configuration Carion et al. [2020]: a 6-layer Transformer encoder and 6-layer decoder with $d_{\text{model}} = 256$, 8 decoder attention heads, and 100 object queries, trained with the standard DETR object detection loss (Hungarian matcher with ℓ_1 and GIoU losses for boxes and cross-entropy for class labels). To study the effect of TQCA in a regime with strong visual features, we replace the default ResNet backbone with frozen vision transformers and feed their tokens into the DETR encoder.

Concretely, we consider two frozen vision encoders:

- **DINOv2-Base** (facebook/dinov2-base) Oquab et al. [2024].
- **DINOv3** (facebook/dinov3-vit7b16-pretrain-lvd1689m) Siméoni et al. [2025].

For each backbone, we resize images to 224×224 , pass them through the frozen vision encoder, and take the resulting patch tokens as input to the DETR encoder. A linear layer maps the backbone embedding dimension to the DETR hidden dimension; during training, this projection, the Transformer encoder-decoder, and the detection head are all trained, while the vision backbone remains frozen.

On the decoder side, we compare:

- **Vanilla DETR decoder.** The standard decoder in which object queries cross-attend to encoder features, followed by residual addition, self-attention, and feed-forward sub-layers.
- **TQCA decoder.** The same decoder, except that each cross-attention sub-layer is replaced by a TQCA layer that applies token mixing on the cross-attended branch before the residual connection; all other components (self-attention, feed-forward networks, layer norms, output head) are unchanged.

We train all models for 100 epochs with AdamW (learning rate 10^{-4} , weight decay 10^{-4}), using linear warmup followed by cosine decay. The vision backbone remains frozen throughout, and we use auxiliary decoder losses following the standard DETR training recipe.

4.2.2 Experiment Results

Table 5 summarizes mAP@0.5 on PASCAL VOC for vanilla DETR and the TQCA decoder across both frozen backbones. In addition to AP, we log two extra diagnostics computed on the validation set: duplicate prediction rate and mean cardinality error, both derived from the matching between predicted and ground-truth boxes at IoU 0.5 and designed to probe typical DETR failure modes beyond AP. TQCA consistently improves detection performance, raising mAP@0.5 from 12.29 ± 0.60 to 18.84 ± 0.95 with DINOv2 and from 13.55 ± 0.70 to 29.50 ± 1.50 with DINOv3, while also improving the structural diagnostics (analyzed in detail below).

Table 5: PASCAL VOC detection with frozen DINO backbones and DETR decoder. We report mAP@0.5 on the validation split and structural diagnostics: duplicate prediction rate and mean cardinality error (mean $\pm$ standard deviation over 5 seeds).

Backbone	Decoder	mAP@0.5 $\uparrow$	Dup. rate $\downarrow$	Card. error (mean) $\downarrow$
DINOv2-Base	Vanilla CA	12.29 ± 0.60	35.16 ± 1.70	83.69 ± 4.00
	TQCA	18.84 ± 0.95	24.38 ± 1.20	76.41 ± 3.80
DINOv3	Vanilla CA	13.55 ± 0.70	33.12 ± 1.65	79.18 ± 3.95
	TQCA	29.50 ± 1.50	12.19 ± 0.60	60.05 ± 3.00

4.2.3 Interpretation and Analysis

We use the duplicate prediction rate and cardinality error to interpret how TQCA reshapes DETR’s decoder behavior. Duplicate prediction rate is computed by matching predictions to ground-truth boxes (class-aware, $\text{IoU} \geq 0.5$) and counting, for each ground-truth instance, how many predictions it attracts; we then average this count across instances. Values close to 1 indicate that on average each

object is covered by a single prediction, while values significantly above 1 signal the classic DETR failure mode where multiple queries latch onto the same object.

Mean cardinality error measures the difference between the number of predictions and the number of ground-truth objects per image, averaged over the validation set. A value near zero means the model tends to predict the correct number of boxes, whereas large positive or negative values reflect systematic over- or under-prediction.

As shown in Table 5, switching from vanilla cross-attention to TQCA consistently improves these structural diagnostics. For DINOv2, the duplicate prediction rate drops from 35.16 ± 1.70 to 24.38 ± 1.20 and the mean cardinality error from 83.69 ± 4.00 to 76.41 ± 3.80 . For the stronger DINOv3 backbone, the duplicate predictions reduced from 33.12 ± 1.65 to 12.19 ± 0.60 and mean cardinality error from 79.18 ± 3.95 to 60.05 ± 3.00 . Taken together, these results suggest how TQCA leads to better-coordinated queries that generate fewer duplicate boxes and a set size closer to the ground-truth objects, therefore contributing to the higher AP.

5 Conclusion

We introduced talking-queries cross-attention (TQCA), which inserts a learnable token-mixing layer on the cross-attended branch before the residual add, directly addressing the dirty residual problem in standard cross-attention blocks. On image–text matching over Flickr30k and COCO, with frozen vision encoders and either frozen or learned text stacks, replacing vanilla cross-attention with TQCA reliably rescued cases where training collapsed to the blind baseline, while behaving essentially neutral when the underlying vision features were already well aligned. We also evaluated TQCA in DETR-style object detection on PASCAL VOC by replacing decoder cross-attention with TQCA under frozen DINOv2 and DINOv3 backbones, which improved mAP@0.5 and reduced duplicate predictions and cardinality error, indicating better coordination among queries. Overall, these results suggest that explicit post-attention token mixing in the pre-add slot is a simple, generally useful modification for stabilizing cross-attention adapters and improving set prediction, and merits further testing in other cross-attention–based systems such as text-conditioned diffusion models.

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A Training Plots

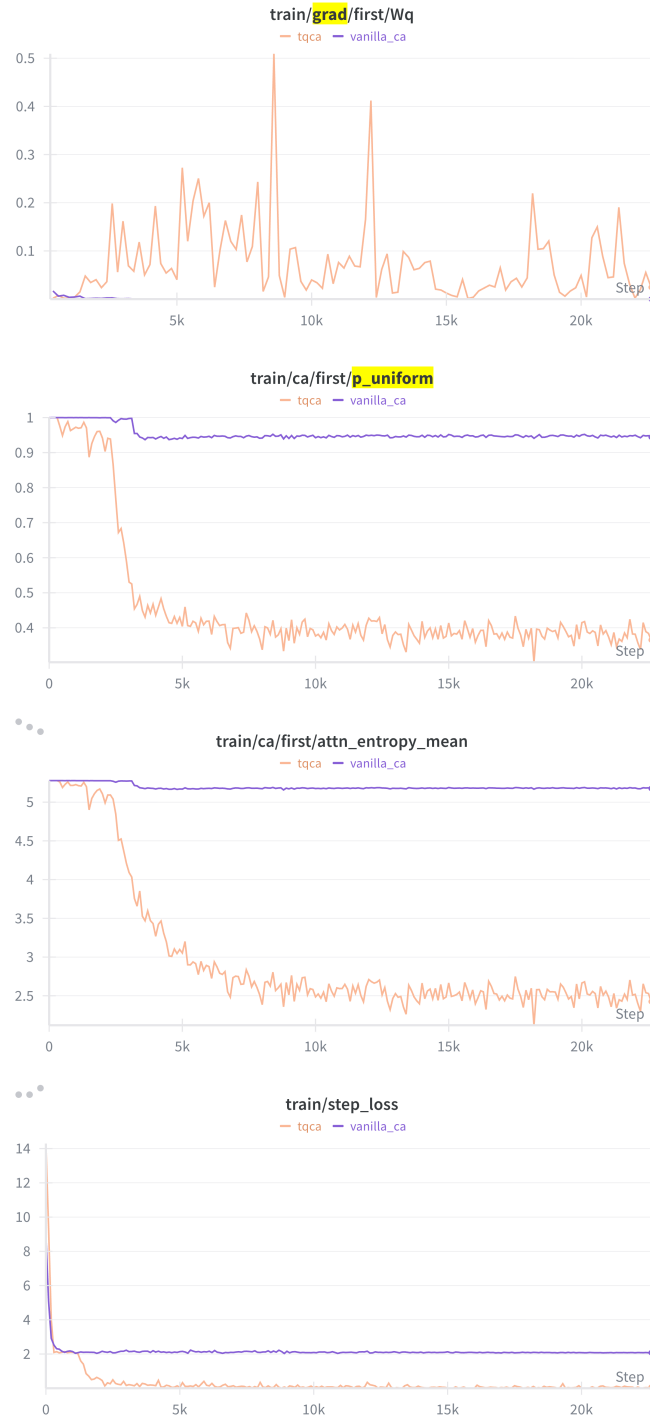


Figure 4: Attention uniformity (p_{uniform}), attention entropy, and training loss over epochs for the CLIP backbone on Flickr30k. TQCA drives attention away from the nearly uniform regime found in the vanilla cross-attention model and achieves stable optimization.